

Note: This document only contains the graded problems viz Exercise 2 and 4. To view my solutions to the full Series 2 problem set, [click here](#).

Exercise 2

Consider three independent random variables $X_1 \sim \text{Uniform}(\theta - 1, \theta)$, $X_2 \sim \text{Uniform}(\theta, \theta + 1)$ and $X_3 \sim \text{Uniform}(\theta + 1, \theta + 2)$. Find a bivariate sufficient statistic for θ based on X_1, X_2, X_3 .

Solution:

We have the following pdf for each X_i

$$f_{X_1}(x_1) = \mathbb{1}_{\{\theta-1 < x_1 < \theta\}} \quad f_{X_2}(x_2) = \mathbb{1}_{\{\theta < x_2 < \theta+1\}} \quad f_{X_3}(x_3) = \mathbb{1}_{\{\theta+1 < x_3 < \theta+2\}}$$

Since X_i 's are independently distributed, we have the following joint pdf

$$\begin{aligned} f_{\tilde{X}}(\tilde{x}) &= \mathbb{1}_{\{\theta-1 < x_1 < \theta\}} \mathbb{1}_{\{\theta < x_2 < \theta+1\}} \mathbb{1}_{\{\theta+1 < x_3 < \theta+2\}} \\ &= \mathbb{1}_{\{x_1 < \theta < x_1+1\}} \mathbb{1}_{\{x_2-1 < \theta < x_2\}} \mathbb{1}_{\{x_3-2 < \theta < x_3-1\}} \\ &= (\mathbb{1}_{\{\theta > x_1\}} \mathbb{1}_{\{\theta > x_2-1\}} \mathbb{1}_{\{\theta > x_3-2\}}) \times (\mathbb{1}_{\{\theta < x_1+1\}} \mathbb{1}_{\{\theta < x_2\}} \mathbb{1}_{\{\theta < x_3-1\}}) \\ &= (\mathbb{1}_{\{\theta > \max\{x_1, x_2-1, x_3-2\}\}}) \times (\mathbb{1}_{\{\theta < \min\{x_1+1, x_2, x_3-1\}\}}) \end{aligned}$$

Now, we'll apply the *Neyman-Fisher Factorization Theorem* to the joint pdf. Thus, we have the following

$$f_{\tilde{X}}(\tilde{x}) = \underbrace{(\mathbb{1}_{\{\theta > \max\{x_1, x_2-1, x_3-2\}\}}) \times (\mathbb{1}_{\{\theta < \min\{x_1+1, x_2, x_3-1\}\}})}_{g_{\theta}(T(\tilde{x}))} \times \underbrace{1}_{h(\tilde{x})}$$

Thus, we have the following

$$g_{\theta}(T(\tilde{x})) = (\mathbb{1}_{\{\theta > \max\{x_1, x_2-1, x_3-2\}\}}) (\mathbb{1}_{\{\theta < \min\{x_1+1, x_2, x_3-1\}\}}) \quad \text{and} \quad h(\tilde{x}) = 1$$

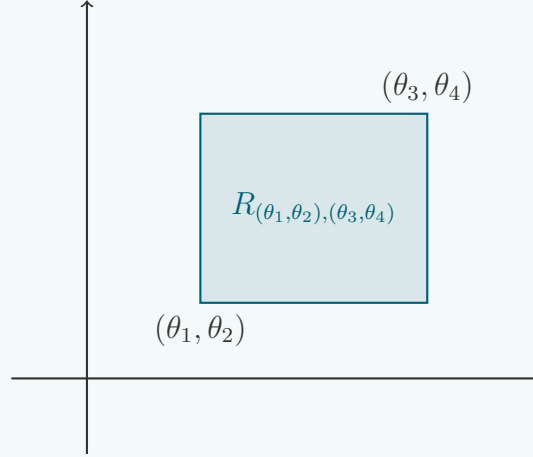
Hence, according to Neyman-Fisher Factorization Theorem,

$$T(\tilde{x}) = (\max\{x_1, x_2 - 1, x_3 - 2\}, \min\{x_1 + 1, x_2, x_3 - 1\})$$

is jointly sufficient for θ

Exercise 4

Let $\theta_1, \theta_2, \theta_3, \theta_4$ be parameters satisfying $\theta_1 < \theta_3$ and $\theta_2 < \theta_4$. Denote by $R_{(\theta_1, \theta_2), (\theta_3, \theta_4)}$ the rectangular region with lower left corner (θ_1, θ_2) and upper right corner (θ_3, θ_4) (see the figure below).



For $\theta = (\theta_1, \theta_2, \theta_3, \theta_4)$, let $f_\theta(x, y)$ be the pdf of the uniform distribution on $R_{(\theta_1, \theta_2), (\theta_3, \theta_4)}$. That is

$$f_\theta(x, y) = \begin{cases} c & \text{if } (x, y) \in R_{(\theta_1, \theta_2), (\theta_3, \theta_4)}, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Determine the value of c .
- (b) Let $(X_1, Y_1), \dots, (X_n, Y_n)$ be i.i.d. bivariate observations with density $f_\theta(x, y)$. Derive the joint pdf of the random sample $(X_1, Y_1), \dots, (X_n, Y_n)$.
- (c) Find a four-dimensional sufficient statistic for $\theta = (\theta_1, \theta_2, \theta_3, \theta_4)$ based on the random sample $(X_1, Y_1), \dots, (X_n, Y_n)$.

Solution:

- (a) First we'll calculate the area, S for the region. We can clearly see that the area is rectangular in nature with the lengths $\theta_3 - \theta_1$ and $\theta_4 - \theta_2$. Thus, the area of a rectangle with the following lengths is $area(S) = (\theta_3 - \theta_1)(\theta_4 - \theta_2)$. Hence, the *bivariate density* of a pair random variable, (X, Y) , distributed uniformly over the area S would be

$$f_\theta(x, y) = \begin{cases} \frac{1}{(\theta_3 - \theta_1)(\theta_4 - \theta_2)} & \text{if } (x, y) \in S \\ 0 & \text{otherwise} \end{cases}$$

Thus, clearly

$$c = \frac{1}{(\theta_3 - \theta_1)(\theta_4 - \theta_2)}$$

(b) Alternatively, the bivariate density can also be written as

$$f_\theta(x, y) = \frac{1}{(\theta_3 - \theta_1)(\theta_4 - \theta_2)} (\mathbb{1}_{\{\theta_1 < x < \theta_3\}} \mathbb{1}_{\{\theta_2 < y < \theta_4\}})$$

Since $(X_1, Y_1), \dots, (X_n, Y_n)$ are independently distributed bivariate, the joint density is just the product of their marginal densities. Hence, we have the following joint density for the given random sample

$$\begin{aligned} f_\theta \left(\underset{\sim}{x}, \underset{\sim}{y} \right) &= \prod_{i=1}^n \frac{1}{(\theta_3 - \theta_1)(\theta_4 - \theta_2)} (\mathbb{1}_{\{\theta_1 < x_i < \theta_3\}} \mathbb{1}_{\{\theta_2 < y_i < \theta_4\}}) \\ &= \left(\frac{1}{(\theta_3 - \theta_1)(\theta_4 - \theta_2)} \right)^n \prod_{i=1}^n (\mathbb{1}_{\{\theta_1 < x_i\}}) (\mathbb{1}_{\{\theta_2 < y_i\}}) (\mathbb{1}_{\{\theta_3 > x_i\}}) (\mathbb{1}_{\{\theta_4 > y_i\}}) \\ &= \left(\frac{1}{(\theta_3 - \theta_1)(\theta_4 - \theta_2)} \right)^n \mathbb{1}_{\{\theta_1 < x_{(1)}\}} \mathbb{1}_{\{\theta_2 < y_{(1)}\}} \mathbb{1}_{\{\theta_3 > x_{(n)}\}} \mathbb{1}_{\{\theta_4 > y_{(n)}\}} \end{aligned}$$

Thus, we have the following result for the joint pdf of n i.i.d bivariate uniformly distributed random variables (X_i, Y_i) in the given area S

$$f_\theta \left(\underset{\sim}{x}, \underset{\sim}{y} \right) = \left(\frac{1}{(\theta_3 - \theta_1)(\theta_4 - \theta_2)} \right)^n \mathbb{1}_{\{\theta_1 < x_{(1)}\}} \mathbb{1}_{\{\theta_2 < y_{(1)}\}} \mathbb{1}_{\{\theta_3 > x_{(n)}\}} \mathbb{1}_{\{\theta_4 > y_{(n)}\}}$$

where $x_{(i)}$ and $y_{(i)}$ are order statistic

(c) To find a sufficient statistic for $\theta = (\theta_1, \theta_2, \theta_3, \theta_4)$, we'll use the Neyman-Fisher Factorization Theorem. Consider the joint pdf in the following form

$$f_\theta \left(\underset{\sim}{x}, \underset{\sim}{y} \right) = \underbrace{\left[\left(\frac{1}{(\theta_3 - \theta_1)(\theta_4 - \theta_2)} \right)^n \mathbb{1}_{\{\theta_1 < x_{(1)}\}} \mathbb{1}_{\{\theta_2 < y_{(1)}\}} \mathbb{1}_{\{\theta_3 > x_{(n)}\}} \mathbb{1}_{\{\theta_4 > y_{(n)}\}} \right]}_{g_\theta \left(\underset{\sim}{x}, \underset{\sim}{y} \right)} \times \underbrace{1}_{h \left(\underset{\sim}{x}, \underset{\sim}{y} \right)}$$

Thus, according to Factorization Theorem, we can conclude that

$$T \left(\underset{\sim}{x}, \underset{\sim}{y} \right) = (x_{(1)}, x_{(n)}, y_{(1)}, y_{(n)})$$

are *jointly sufficient* for θ , where $x_{(i)}$ and $y_{(i)}$ are order statistic